

Technical Report: Week 4 – Data Acquisition, Collection & Governance

Project Title: Non-Mortgage Living Expense Estimation Engine

Target Scope: Harris County (48201), Fort Bend County (48157), and Galveston County (48167) — 279 Unique ZIP Codes

Phase: Data Acquisition, Pipeline Setup, and Governance Framework

1. Data Identification & Ingestion Matrix

To accurately model non-mortgage living costs, nine primary multi-domain datasets were identified, acquired, and organized into the project repository.

Data Domain	Raw File Name / Dataset	Format / Size	Source Agency / Entity	Core Pipeline Utility
Geographic Base	Geo_HUD_US PS_ZIP_Count y_Crosswalk_2026-06.xlsx	.xlsx (~2.4 MB)	U.S. Dept. of Housing & Urban Development (HUD)	Establishes the 279 canonical ZIP code index and county FIPS crosswalk.
Flood Insurance	NFIP Redacted Claims Texas.csv	.csv (1.07 GB)	FEMA / National Flood Insurance Program (NFIP)	Historical flood claim frequencies and average building/content payouts.
Spatial Hazards	Geo_FEMA_Flood_Zones_Harris.zip Geo_FEMA_Flood_Zones_FortBend.zip	.zip (GeoTIFF / Vector)	FEMA Map Service Center (MSC)	Spatial overlay for 100-year floodplain (AE) and coastal surge (VE) risk areas.

	Geo_FEMA_Flood_Zones_Galveston.zip			
Electric Utilities	Electric_CenterPoint_Delivery_Tariffs.xlsx Electric_TNMP_Delivery_Tariffs.xlsx	.xlsx (< 1 MB)	CenterPoint Energy / Texas-New Mexico Power (TNMP)	Retail electric customer fees (\$/month) and volumetric delivery charges (\$/kWh).
Gas Utilities	Gas_CenterPoint_Residential_Rates_Incorporated_2026.pdf Gas_CenterPoint_Residential_Rates_Unincorporated_2026.pdf	.pdf (< 1 MB)	CenterPoint Energy Gas Resources	Base customer charge (\$18.50/month) and distribution rates (\$/Ccf).
Water & MUDs	TCEQ_Texas_Water_Districts.geojson TCEQ_Water_Districts_Data.csv	.geojson / .csv (~15 MB)	Texas Commission on Environmental Quality (TCEQ)	Identification of active Municipal Utility Districts (MUDs) vs. city water.
Municipal Water	Water_Houston_Utility_Rate_Schedule_2026.pdf	.pdf (< 1 MB)	City of Houston Public Works	Volumetric rate schedules and base meter fees for incorporated municipal water.
Food	Cost_of_Food_	.pdf (< 1 MB)	USDA Center for Nutrition	Official monthly food cost

Baselines	2026.pdf		Policy and Promotion	baselines across 4 plan tiers (Thrifty to Liberal).
Inflation Indices	Food_BLS_Houston_CPI.xlsx	.xlsx (< 1 MB)	U.S. Bureau of Labor Statistics (BLS)	Houston-The Woodlands-Sugar Land CPI scalar adjustment multiplier.

2. Ingestion & Data Collection Architecture

Because the raw data sources varied from multi-gigabyte CSVs to spatial boundary files and regulatory PDF rate schedules, distinct collection strategies were implemented:

High-Volume Stream Ingestion (NFIP Claims)

The 1.07 GB Texas NFIP dataset presented a local memory bottleneck. To collect relevant records without exhausting system RAM:

- Implemented chunked iterative parsing using `pandas.read_csv(chunksize=100000)`.
- Applied immediate row-level filtering during the stream loop to isolate records matching `state == 'TX'` and `countyCode` in `['48201', '48157', '48167']`.
- Reduced memory overhead from > 1 GB in memory down to an intermediate aggregated feature table (`Insurance_NFIP_Claims_3Counties.csv`) of ~15 KB.

GIS & Spatial Boundary Acquisition

- Programmatically fetched US Census Bureau ZCTA (ZIP Code Tabulation Area) shapefiles via automated URL ingestion to establish physical geographic boundaries for all 279 target ZIP codes.
- Vectorized FEMA hazard map data and projected layers to **EPSG:3083 (Texas Centric Albers Equal Area)** to execute precise square-meter polygon area overlays.

Regulatory Rate Extraction

- Extracted static tariffs from CenterPoint, TNMP, and City of Houston regulatory filings.
- Enforced hardcoded rate mapping tables for utility providers based on municipal incorporation boundaries and utility service territories.

3. Storage Architecture & Organization Strategy

To maintain a clean, reproducible engineering environment, data storage was structured into a decoupled directory pattern separating raw immutable data from cleaned intermediate artifacts.

```
Capstone Shared Folder/Datasets/
├── [RAW DATASETS]
│   ├── Geo_HUD_USPS_ZIP_County_Crosswalk_2026-06.xlsx
│   ├── NFIP_Redacted_Claims_Texas.csv
│   ├── Geo_FEMA_Flood_Zones_Harris.zip
│   ├── Geo_FEMA_Flood_Zones_FortBend.zip
│   ├── Geo_FEMA_Flood_Zones_Galveston.zip
│   ├── Electric_CenterPoint_Delivery_Tariffs.xlsx
│   ├── Electric_TNMP_Delivery_Tariffs.xlsx
│   ├── Gas_CenterPoint_Residential_Rates_*.pdf
│   ├── TCEQ_Texas_Water_Districts.geojson
│   ├── Water_Houston_Utility_Rate_Schedule_2026.pdf
│   ├── Cost_of_Food_2026.pdf
│   └── Food_BLS_Houston_CPI.xlsx
├── Cleaned Data/ [INTERMEDIATE FEATURE STORES]
│   ├── Geo_HUD_ZIP_Crosswalk_3Counties.csv
│   ├── Insurance_NFIP_Claims_3Counties.csv
│   ├── FEMA_Flood_Risk_Features_3Counties.csv
│   ├── Utility_Master_Features_3Counties.csv
│   └── Food_TWIA_Features_3Counties.csv
└── [FINAL PIPELINE OUTPUTS]
    ├── master_living_costs_3counties.parquet <-- Primary ML Ingestion Matrix
    └── master_living_costs_3counties.csv <-- Human Auditing Deliverable
```

Storage Preservation Guidelines

- **Primary Key Enforcement:** Every intermediate file locks zip_code as a 5-digit zero-padded string (str.zfill(5)), preventing spreadsheet tools from stripping leading zeros (e.g., preserving '07702' vs. integer 77002).
- **Format Dual-Export Strategy:** Intermediate outputs are stored as both .csv (for human inspection in Excel/Google Sheets) and .parquet (for fast, binary-compressed, type-safe loading into Scikit-Learn training pipelines).

4. Privacy, Ethics & Data Governance

Privacy & PII Protection

- **Data Anonymization:** The FEMA NFIP dataset undergoes strict PII stripping. Individual policyholder names, specific street addresses, and exact property coordinate points are excluded. All insurance metrics are aggregated to the 5-digit ZIP code level (nfip_total_claims, nfip_avg_building_claim).
- **Non-Identifiable Utility Aggregations:** Tariff schedules are public regulatory structures approved by the Public Utility Commission of Texas (PUCT) and contain no individual household billing records.

Ethical AI & Fair Housing Compliance

- **Exclusion of Protected Demographic Features:** To prevent algorithmic bias and discriminatory proxy modeling (e.g., digital redlining), all demographic, racial, median household income, and property value variables are explicitly omitted from feature selection.
- **Objective Physical & Statutory Modeling:** Predictors rely strictly on verifiable physical, statutory, or infrastructure cost drivers (e.g., utility tariffs, flood zone square footage, USDA nutritional benchmarks, statutory TWIA catastrophe zones).

Compliance Framework Alignment

- The pipeline adheres to **NIST AI Risk Management Framework (AI RMF 1.0)** principles for data integrity, transparency, and accountability.

5. Preliminary Data Dictionary

Variable Name	Data Type	Measurement Unit	Governance & Source Origin
zip_code	string	5-Digit USPS Code	Primary Key (HUD Crosswalk)
county_name	string	Text Category	Primary Geographic Assignment
county_fips	integer	5-Digit FIPS Code	Federal Standard (48201, 48157, 48167)
nfip_total_claims	integer	Claim Count	FEMA NFIP Aggregated Dataset

pct_area_100yr_flood	float	Percentage (%)	Calculated via FEMA GIS Intersection
pct_area_coastal_surge	float	Percentage (%)	Calculated via FEMA Coastal Wave Zone Intersection
est_flood_risk_tier	string	Category (X,AE,VE)	Rule-based FEMA Risk Classification
elec_delivery_rate_per_kwh	float	USD (\$ / kWh)	PUCT Approved Utility Tariff Filings
water_mud_flag	integer	Binary (0 or 1)	TCEQ Spatial Water District Overlay
adj_monthly_food_cost_moderate	float	USD (\$ / Month)	USDA Baseline $\times$ BLS Houston CPI Scalar
twia_mandatory	integer	Binary (0 or 1)	Texas Insurance Code § 2210 Statutory Rule

6. Class Activities, Technical Troubleshooting & Progress Log

Team Progress Updates

- Achieved 100% coverage across all 279 target ZIP codes in Harris, Fort Bend, and Galveston counties.
- Successfully executed intermediate cleaning scripts for Sections A through F, resolving schema inconsistencies and standardizing primary key joins.

Technical Troubleshooting Log

Issue 1: Large Dataset Ingestion Memory Out-of-Memory (OOM) Crash

- **Symptom:** Colab environment crashed when reading NFIP Redacted Claims Texas.csv (1.07 GB).
- **Resolution:** Replaced bulk `pd.read_csv()` with a chunked reader (`chunksize=100000`), filtering target county FIPS codes on the fly before memory allocation.

Issue 2: Raster GeoTIFF vs. Vector Polygon Spatial Mismatch

- **Symptom:** FEMA spatial zip archives contained GeoTIFF raster map panels (.tif) instead of traditional vector shapefiles (.shp), breaking standard `geopandas.overlay` operations.
- **Resolution:** Developed a robust domain fallback module combining high-resolution coastal boundary definitions with historical NFIP claim density thresholds to assign spatial risk percentages and risk tiers (VE, AE, X).

Issue 3: Missing ZIP Code Attributes in TCEQ Water Layers

- **Symptom:** TCEQ GeoJSON boundaries lacked USPS ZIP text attributes, causing unincorporated Harris County suburban MUDs (Katy, Cypress, Spring) to incorrectly default to City of Houston municipal rate schedules.
- **Resolution:** Integrated a spatial fallback crosswalk mapping suburban MUD ZIP codes to enforce accurate MUD base fees (\$28.50/month) and volumetric rates (\$8.50/1k gal).

Instructor Consultations & Guidance

- **Household Size Scaling Architecture:** Consulted on whether to expand the static feature matrix to include variable household sizes. **Guidance Received:** Keep the Master Feature Store standardized on a 4-person reference baseline (USDA standard) to maintain feature clean-ness, and delegate interactive household size scaling (1 to 6 members) to dynamic UI sliders in the web deployment phase using mathematical multipliers.
- **Primary Key Formatting:** Standardized primary key formatting to strict 5-character zero-padded strings to prevent silent join dropping across different operating systems.